Multibranch Fusion-Based Feature Enhance for Remote-Sensing Scene Classification

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***Abstract*— Remote-sensing (RS) scene classification is a fundamental and significant task in RS image interpretation, involving the annotation of semantic content. RS scene images are characterized by complex backgrounds, rich content, and multiscale targets, exhibiting both intraclass separation and interclass convergence. Therefore, extracting features that effectively express the intrinsic attributes of images and possess high discriminative is crucial for RS scene classification. Existing global-based methods often lack the ability to capture significant detailed information in similar scenes. Conversely, methods based on local discriminative features tend to overlook the interrelationships of objects within the same scene. To address these issues, this article proposes a unified framework named MBFNet to align and fuse features of different scales and levels for accurate RS scene classification. We utilize a multibranch feature-extracting network structure with parallel convolution and Transformer modules. Simultaneously, a kernel-selected multiscale aggregation (KSMSA) module is designed to efficiently process the diverse scale features emanating from these parallel branches. By selecting different convolution kernels, a dynamic receptive field is established to adaptively process features of different scales, reducing semantic differences to achieve effective aggregation of multiscale features. Moreover, a learnable multilevel aggregation (LMLA) module is designed to integrate shallow features, such as shape information, into deep features for more comprehensive feature fusion. Benefiting from KSMSA and LMLA, the proposed MBFNet improves the discriminability of features, thereby enhancing classification performance. Comprehensive experiments on three benchmark datasets demonstrate that the proposed method outperforms state-of-the-art RS scene classification methods in terms of performance.**

***Index Terms*— Feature fusion, multilevel aggregation, multiscale aggregation, remote-sensing (RS), scene classification.**

Received 31 October 2024; revised 16 December 2024 and 24 January 2025; accepted 19 February 2025. Date of publication 28 February 2025; date of current version 18 March 2025. This work was supported in part by Hainan Province "Nanhai New Star" Technology Innovation Talent Platform Project under Grant NHXXRCXM202361; in part by the Youth Fund Project of Hainan Natural Science Foundation under Grant 622QN344; in part by the National Key Research and Development Program of China under Grant 2022ZD0160604; in part by NSFC under Grant 62176194; in part by the Key Research and Development Program of Hubei Province under Grant 2023BAB083; in part by the Project of Sanya Yazhou Bay Science and Technology City under Grant SCKJ-JYRC-2022-76 and Grant SKJC-2022-PTDX-031; in part by the Major Project of IoV under Grant 2020AAA001; in part by the Project of Sanya Science and Education Innovation Park, Wuhan University of Technology, under Grant 2021KF0031 and Grant 2022KF0020; in part by MindSpore, which is a new deep learning computing framework (https://www.mindspore.cn/); in part by the National Natural Science Foundation of China under Grant 62271484; in part by the National Science Fund for Distinguished Young Scholars under Grant 61925112; in part by the Key Research and Development Program of Shaanxi under Grant 2023-YBGY225; in part by the High Performance Computing Platform of YZBSTCACC; and in part by CSTC under Grant cstc2021jcyj-msxmX1148. *(Corresponding authors: Yaxiong Chen; Shengwu Xiong.)*

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Digital Object Identifier 10.1109/TGRS.2025.3546638

I. Introduction

As a fundamental and significant task in remote-sensing (RS) image interpretation, RS scene classification involves mapping images' scene content into semantic labels. It plays a vital role in many real-life applications, for instance, in land use [1], urban planning [2], precision agriculture [3], and climate change response [4]. However, RS scene classification still faces two main challenges: the scale differences present in the image and the level differences in feature extraction. In the first, RS scene images, due to the influence of different factors such as the angle of photography and the height from the ground by aerial devices, present objects of varying proportions in the images. As depicted in Fig. 1, scenes or objects containing small-scale images may overlook some details, leading to information loss, while large-scale images encompass more areas, resulting in information redundancy. In the second, the deep learning method is represented by convolutional neural networks (CNNs), which usually only use the features from the last layer for classification. As shown in Fig. 2, the deep features are rich in semantic information, while shallow features reflect the details and shape information of the related objects. Only relying on the last layer for classification tends to overlook low-level features, which are also of great importance for RS scene classification.

To alleviate the aforementioned issues, some works [6], [7], [8], [9] have expanded CNNs by building multibranch structures that operate on different layers to achieve more comprehensive feature extraction. By focusing on different feature information through multiple branches and utilizing various aggregation strategies, the representation ability of features is enhanced. For instance, Wang et al. [8] proposes a global–local dual-stream architecture that extracts global and local features separately from the entire image and the

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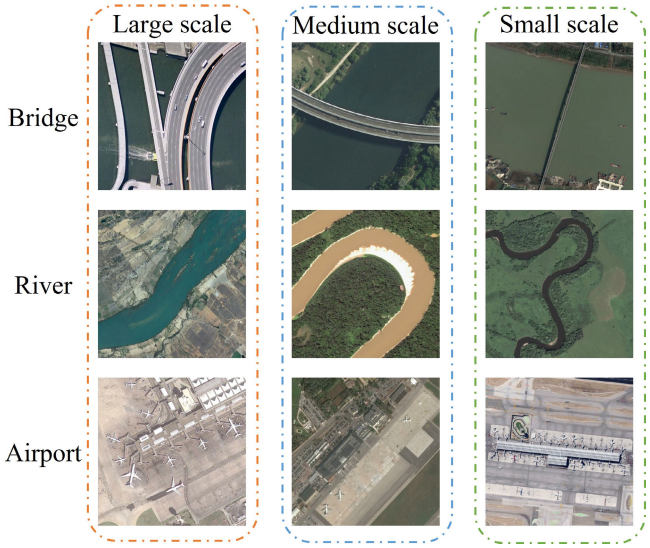


Fig. 1. Large-, medium-, and small-scale images of the "Bridge," "River," and "Airport" scenes, selected from the AID dataset [5]. Specifically, at a large scale, the structure and shape of the main object are clearly visible, whereas at a small scale, the main object only occupies a few pixels.

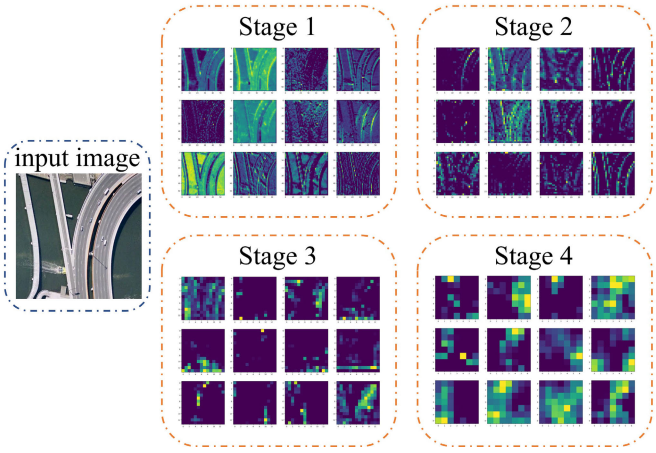


Fig. 2. Feature map of a bridge scene image, extracted from various hierarchical stages of a convolutional network (ResNet50 [19]), contains distinct characteristic information at each level. The low-level convolutional layers focus on capturing linear and edge features, the mid-level convolutional layers emphasize object characteristics, and the high-level convolutional layers tend to describe the overall information of the scene.

most important regions. A structured key region localization module connects the two branches to aggregate features for scene classification. Given that convolutions tend to extract local features of images, the attention mechanism [10] is also employed to fuse with CNN features to improve classification performance [11], [12], [13], [14]. For example, Bai et al. [13] introduce a novel lightweight multiscale deep learning network with efficient spatial pyramid attention, which extends the receptive field through multiscale depth convolutions and adopts different scale feature fusion methods at various levels. This approach enhances the network's ability to capture and integrate features across different scales, thereby improving classification accuracy.

More recently, methods that integrate both local features extracted by CNNs and global features extracted by Transformers have been proposed [15], [16], [17], [18]. For instance, Tang et al. [18] present the efficient multiscale transformer and cross-level attention learning (EMTCAL) model, which employs a multilayer feature extraction module to obtain global visual features and multilevel convolutional features from RS scenes. However, a critical aspect often overlooked in the fusion of these two structural branches is the inherent differences between them. Due to the significant semantic span of features across different scales, direct fusion can introduce semantic discrepancies, resulting in redundant information within the fused features, which may negatively impact classification performance.

In this article, we present a unified framework named MBFNet, which integrates the ability of CNNs to capture local details with Transformers' proficiency in modeling long-range dependencies and capturing global contextual information. To achieve this, the framework employs a multibranch design, where each branch is specifically designed to process information at different scales. The MBFNet mainly contains two novel modules: a kernel-selected multiscale aggregation (KSMSA) module and a learnable multilevel aggregation (LMLA) module. The proposed KSMSA module applies different sizes of convolution kernels to process features at various scales through dynamic convolution kernel selection, aligning semantic differences and effectively capturing diverse scale information. This enables better aggregation of features from the different branches and the acquisition of comprehensive scale information. The LMLA module processes shallow information and dynamically integrates it into deep features, addressing the lack of shape and edge details in deep features, enhancing feature representation, and improving classification accuracy.

By investigating the above ideas, the main contributions of our work can be summarized as follows.

1) We propose a unified framework named MBFNet that aligns and fuses features of different scales and levels for accurate RS scene classification.
2) We introduce the KSMSA module, which allows for better aggregation of features from different branches and the acquisition of comprehensive scale information.
3) We develop the LMLA module to process shallow information and dynamically integrate it into deep features, enhancing the overall feature representation.
4) Comprehensive experiments on three benchmark datasets demonstrate the effectiveness of the proposed MBFNet method for RS scene classification.

The rest of this article is organized as follows. First, in Section II, we provide a comprehensive overview of relevant methods for RS scene classification. Next, in Section III, we elaborate on the network architecture of the proposed MBFNet. Then, in Section IV, we detail the experimental process and conduct a comparative analysis. Finally, in Section V, we summarize our work.

II. Related Works

A. CNN-Based RS Scene Classification

In the early stages, manually designed features were applied to RS image classification. Methods such as scale-invariant

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feature transform (SIFT) [20], histogram of oriented gradients (HOG) [21], and spectral indices of ground objects [22] relied on handcrafted descriptive features to represent RS scenes. Subsequently, some works aimed to learn a set of basis functions for image feature encoding that could represent certain semantic information. For instance, methods such as the bag-of-visual-words model [23], probabilistic topic models [24], vector of locally aggregated descriptors (VLAD) [25], and Fisher kernels [26] were employed to extract more global information. However, these methods inherently focused on relatively simple shallow features, unable to fully capture the high-level semantic information contained in RS images, resulting in weaker feature representation capabilities. Additionally, designing well-performing manually crafted features is a time-consuming and domain-knowledge-intensive process, further limiting their practical application.

Deep neural networks (DNNs), particularly CNNs, have successfully broken the limitations of traditional hand-crafted features with their remarkable extraction capabilities. Existing CNN-based methods for RS scene classification can be broadly categorized into three groups. The first group focuses on extracting global features. For instance, Li et al. [27] utilized transfer learning in CNNs to learn robust feature representations, thereby enhancing classification accuracy with limited labeled samples. Xie et al. [28] proposed a scale-free CNN that processes RS images of arbitrary sizes by replacing the fully connected (FC) layer with a convolutional layer. This approach overcomes the limitation of the fixed input size required by pretrained networks and preserves critical information during image scaling. Additionally, Zhang et al. [29] introduced an attribute-cooperated CNN, which uses acquired attributes as supplementary information to extract discriminative visual features for RS scene classification.

RS scene classification tends to categorize the overall events in an image rather than focusing on specific objects. Consequently, the second category of approaches aims to aggregate multilayer features. For example, Sun et al. [7] introduced the gated bidirectional network (GBNet), a unified framework that seamlessly integrates hierarchical feature aggregation with noise reduction, thereby efficiently amalgamating multilayer features. Similarly, Xu et al. [30] presented a framework leveraging graph CNNs (GCNNs) to aggregate deep features, comprehensively capturing and fusing global features alongside multilayer contextual information, ultimately enhancing classification accuracy. Furthermore, the feature pyramid architecture has garnered widespread adoption due to its robustness in managing multilevel features. Wang et al. [31] devised an enhanced deep semantic embedding (DSE) pyramid network tailored for RS images. This network excels at extracting multilevel features and incorporates a dual-branch deep feature fusion module to integrate these features, fostering complementary strengths and bolstering classification precision. Meanwhile, Li et al. [32] proposed the random topology and random multiscale mapping (RTRMM) method, a multiscale lightweight architecture tailored for RS image recognition. Guided by mathematical analysis, they crafted a variable search space comprising decomposable convolutional units, where each neuron's mapping is dictated by a random multiscale mapping sampler. Ultimately, these topologies and mappings are orchestrated into cohesive blocks.

The final group of approaches involves integrating CNNs with other technologies to harness their unique strengths and enhance RS scene classification performance. Zhang et al. [33] incorporated capsule networks (CapsNet) into CNNs to encode both attribute and spatial information in RS images, thereby improving classification accuracy. Shawky et al. [34] proposed a data augmentation strategy based on a CNN-MLP architecture, utilizing the Xception network as a feature extractor to fully leverage the advantages of both CNNs and MLPs. Additionally, by integrating a metric learning mechanism, Cheng et al. [35] enhanced the capture of discriminative features, achieving notable performance with high-computational efficiency.

B. Attention-Based RS Scene Classification

Despite the significant performance advantages of CNN-based methods in RS scene classification, they still encounter notable challenges in capturing global contextual information. Traditional convolutional modules are limited by their local receptive fields, primarily focusing on short-range dependencies within hierarchical structures. This often results in difficulties in effectively capturing long-range dependencies, which are crucial for RS scene classification tasks. To address this limitation, visual attention technology has been progressively introduced into RS image processing, leading to the development of innovative RS scene classification methods. Based on the evolution of attention mechanisms, these methods can be broadly classified into two categories: traditional-attention-based methods, which employ meticulously designed attention models to enhance the representation of key regions and improve classification performance; and Transformer-based methods, which utilize the powerful sequence modeling capabilities of Transformers to capture dependencies over longer spatial distances, thereby offering new perspectives and solutions for RS scene classification.

Traditional-attention-based methods are effective at selectively learning important information to improve model performance. For instance, Zhao et al. [36] designed an enhanced attention module (EAM) to improve the extraction of discriminative features. This EAM utilizes two dilated convolution branches, each with a different dilation rate, to expand the receptive field without increasing the number of parameters (Params). Fu et al. [37] proposed a recurrent economical attention module that captures global contextual information while reducing model Params. In contrast, Tang et al. [38] introduced an attention consistency network (ACNet) for RS scene classification, which integrates spatial and channel attention mechanisms to extract critical features from images.

The second group is based on the Transformer framework. With the emergence of the vision transformer (ViT) [39], which has demonstrated exceptional performance in capturing contextual features, there has been significant advancement in Transformer-based models. The success of the ViT model on general datasets has spurred the development of various

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Transformer variants. Liu et al. [40] introduced the Swin Transformer, which is designed for diverse visual tasks and reduces the computational cost associated with the self-attention mechanism. Building on CNN design principles, Heo et al. [41] developed the pooling-based vision transformer (PiT), exploring the role of spatial dimension transformation and its efficacy within the Transformer framework. Yuan et al. [42] proposed the token-to-token vision transformer (T2T-ViT), which incorporates a novel layer-wise Token-To-Token transformation. This approach progressively structures images into tokens by recursively aggregating adjacent tokens.

However, the calculation of attention exhibits square complexity. A special linear attention transformer, known as Mamba [43], [44], has been proposed, featuring approximate linear complexity. Due to its exceptional ability to model long-range dependencies [45], Mamba-based methods have been widely adopted in the field of remote sensing [46], [47], [48]. Recent studies suggest that Mamba-based methods are more suited to tasks involving long sequences and autoregressive properties [49]. Since RS scene classification tasks lack autoregressive attributes, this article still adopts a more traditional transformer approach.

In the field of RS scene classification, Zhang et al. [17] introduced the RS Transformer, a model designed to capture global contextual information by integrating self-attention mechanisms with ResNet [19], thereby replacing the traditional 3×3 spatial convolutions in the bottleneck layers. Bashmal et al. [50] proposed a Transformer-based approach leveraging the original Transformer model to effectively capture contextual information in RS imagery, achieving notable classification accuracy. Additionally, Tang et al. [18] developed the EMTCAL model. This model employs multilayer feature extraction modules to acquire both global visual and multilevel convolutional features from RS scenes, and incorporates a context information extraction module to capture intricate contextual details from these multilevel features.

C. Hybrid Architectures-Based RS Scene Classification

Models leveraging conventional monolithic frameworks face significant challenges when processing increasingly intricate and diversified datasets in the field of RS. Consequently, there is a need for models that incorporate hybrid architectures. Hybrid architectures demonstrate the ability to holistically leverage the strengths of distinct models by integrating feature representations across various hierarchical levels. This integration not only facilitates the acquisition of a more extensive and nuanced array of feature information but also ultimately enhances the model's generalization capability. For example, Ma et al. [51] developed an innovative adaptive hybrid fusion network specifically tailored for multiresolution RS image classification. Chen et al. [52] introduced the sparse token transformers (STT) and integrated these modules with CNNs for building extraction without increasing computational complexity or space consumption. Zhu et al. [53] proposed a multisource heterogeneous data fusion framework to mine multidimensional information from multisource heterogeneous data for local climate zone mapping.

The hybrid architecture has also been introduced into RS scene classification to enhance the model's performance. For instance, Chen et al. [54] proposed a hierarchical feature fusion of transformer with patch dilating to capture rich contextual information from hierarchical features. Shen et al. [6] integrated both the ResNet-50 and DenseNet-121 models within a unified CNN framework to comprehensively capture hierarchical features spanning from low to high levels, thereby enabling the extraction of a more enriched feature representation. Guo et al. [55] formulated a sophisticated multiview feature learning framework, encompassing three integral branches: a channel-spatial attention mechanism branch for identifying and emphasizing discriminative feature zones, a triplet metric learning branch, and a center-based metric learning branch. The latter two branches are collectively aimed at enhancing interclass separability and mitigating intraclass variability. Chen et al. [11] incorporated a multibranch local attention network within the context of their architecture, presenting a novel local attention module designed to efficiently capture both channel-wise and spatial attention weights.

Despite significant advancements achieved by the aforementioned methods, a pivotal yet often overlooked aspect of fusing features from hybrid architectures is their inherent disparities. The substantial semantic divergence among features of different scales and levels can lead to semantic inconsistencies when fused directly, resulting in redundant information within the fused features and adversely impacting classification performance. Therefore, this article explores dynamic and learnable feature fusion by implementing the KSMSA and LMLA modules.

III. Methodology

We propose a novel multibranch fusion-based feature enhancement model named MBFNet, which integrates CNN and Transformer architectures, extending the MPViT [15] network. This model employs multiple parallel branches at each layer to process features across different scales and incorporates a novel KSMSA module to effectively aggregate these multiscale features. Additionally, it integrates an LMLA module to enhance deep-level features with supplementary shallow-level information. Detailed descriptions are provided in Sections III-A–III-C.

A. Network Architecture

The framework of the proposed MBFNet is illustrated in Fig. 3, adopting a pyramid-like structure of CNN and divided into four stages (the resolution is reduced to half of the previous stage at each stage), aiming to process feature maps of different scales. Each stage comprises a multiscale embedding module (MSEM), a multibranch module (MBM), and KSMSA. Finally, features from different levels are aggregated through the LMLA module. Due to the stem convolution module can better embed low-level feature representations without losing important information. Therefore, this module is first used to extract the embedded features of input image $x \in \mathbb{R}^{3\times H\times W}$. Specifically, the stem convention module consists of two 3×3 conventional layers with a channel size

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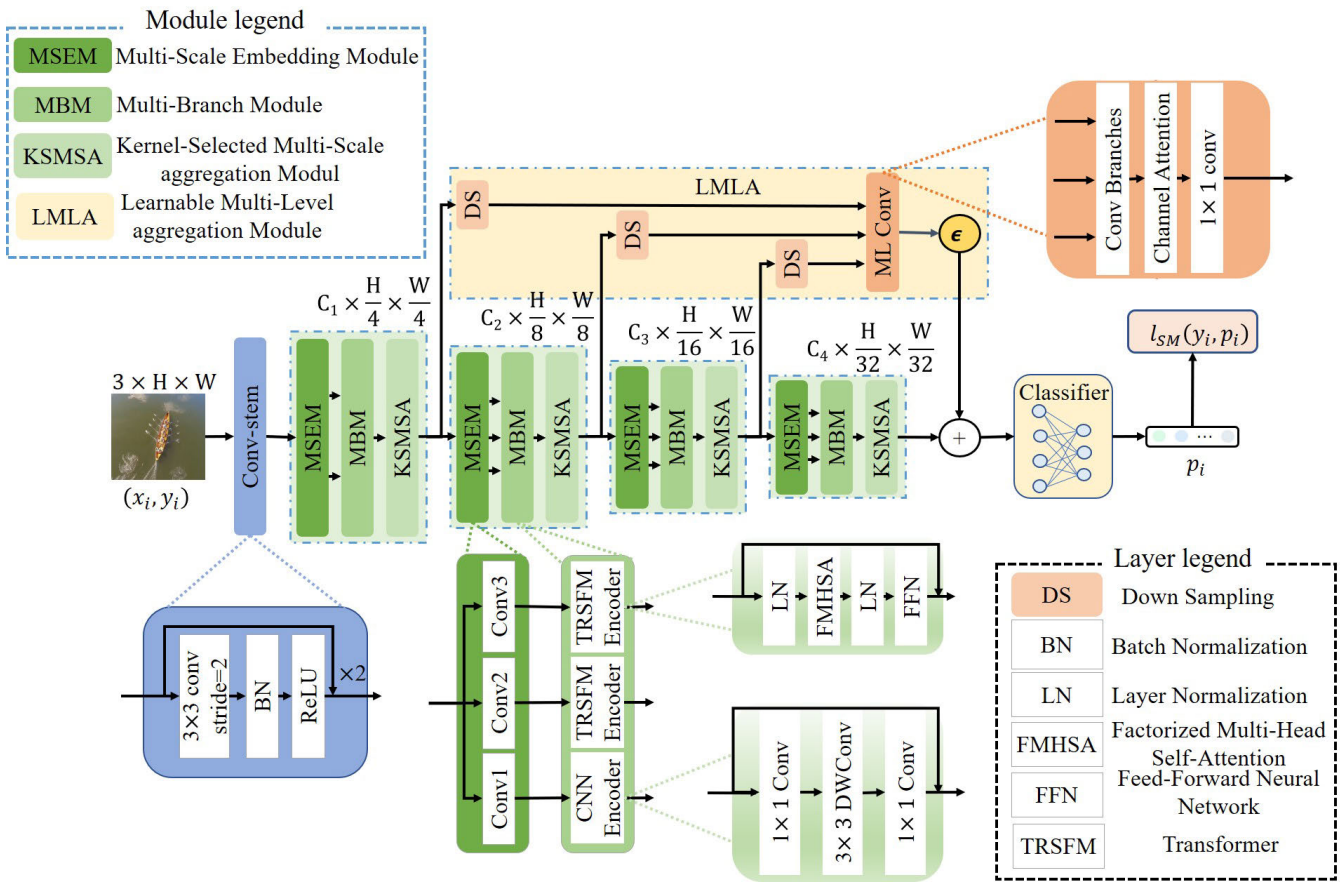


Fig. 3. Framework of the proposed MBFNet adopts a pyramid-like structure of CNNs and is divided into four stages, aiming to process feature maps of different scales and levels.

of $C_1/2$ and C_1 and a stride of 2, the shape of generated feature of $C_1 \times (H/4) \times (H/4)$, where C_1 represents the input dimension of the embedding module in the first stage.

The MSEM adjusts the sequence length of tokens by modifying the stride and padding in convolutional computations, thereby producing features of different scales using various patch sizes (kernel sizes). Furthermore, consecutively stacking convolutional operations with the same channels and filter sizes can expand the receptive field and require fewer Params. To minimize model Params and computational costs, this process employs 3×3 depthwise separable convolutional blocks. Each block consists of a 3×3 depthwise convolution and a 1×1 pointwise convolution within the embedding layer. Batch normalization and the Hardswish activation function are applied following all convolutional layers. Ultimately, the token embedding features of different sizes are independently fed into the encoders of each branch. The four stages feature branch counts of 2, 3, 3, and 3, respectively. Specifically, stages with two branches utilize convolutions of 3×3 and 5×5 (two 3×3), while those with 3 branches incorporate an extra 7×7 (three 3×3) convolutional feature.

In the MBM, a deep residual bottleneck layer is employed to process the embedded features and acquire local feature information. This layer is composed of a 1×1 convolution, a 3×3 depthwise convolution, another 1×1 convolution, and residual connections. Furthermore, the RS scene classification task requires extracting more global information at different scales, and the Transformer possesses excellent global feature extraction capabilities. Hence, the multiscale features from the MSEM are input into a multibranch Transformer encoder. Each Transformer branch adheres to the classical Transformer encoder architecture. However, to mitigate the computational burden introduced by the multibranch approach, the original attention module is modified by adopting an efficient factorized multihead self-attention (FMHSA) mechanism combined with convolutional relative position encoding, forming a novel convolutional attention module.

After the MBM, local features extracted by convolutions and multiscale features processed by Transformers are obtained. Subsequently, the proposed KSMSA module is utilized to effectively aggregate features of different scales from both CNNs and Transformers. Following feature extraction at various stages, to enrich the feature representation, the features from preceding stages—namely, the shallow features from the CNNs—are effectively summed into the final feature representation using an LMLA module. This approach enhances the overall feature representation capability.

B. Kernel-Selected Multiscale Aggregation

The MBM utilizes multiple different branches to extract multiscale features, which have significant semantic differences. In the RS scene classification task, it is crucial to pay attention to objects of multiple scales and the interrelationships

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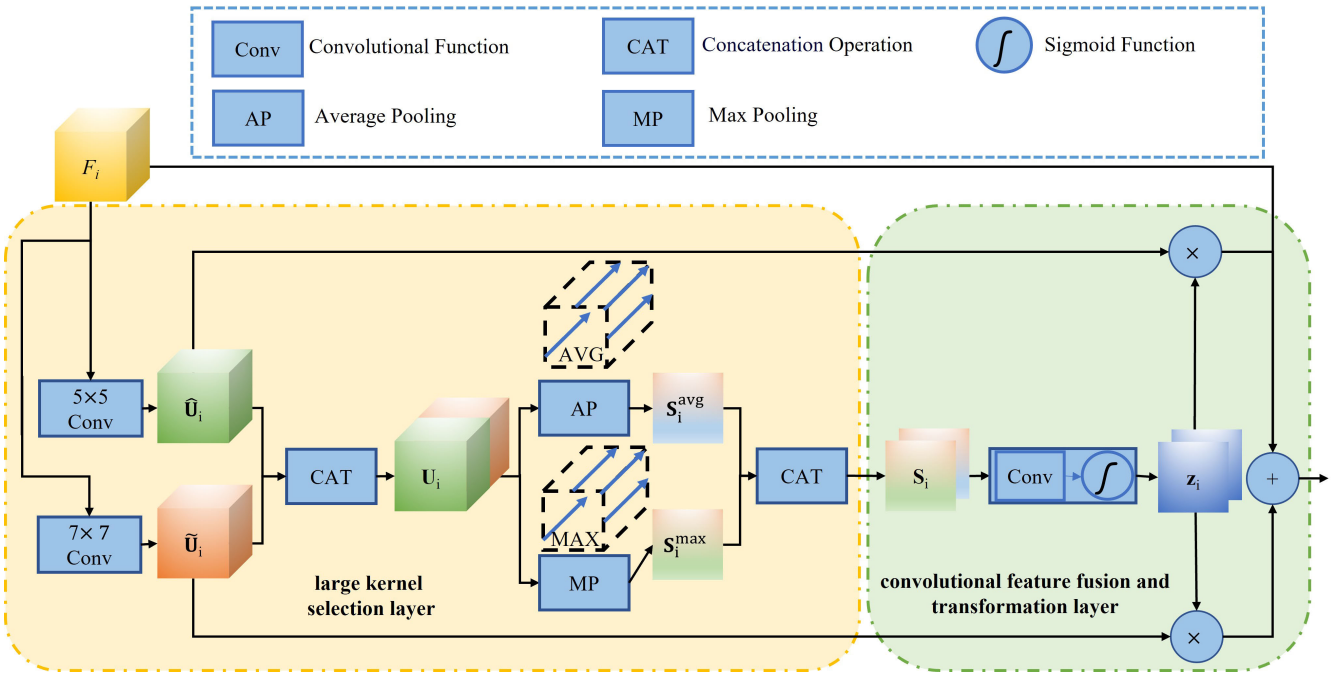


Fig. 4. Structure of proposed KSMSA module, which mainly composed of a large kernel selection and a convolutional feature fusion and transformation layer.

between them. If direct fusion is achieved through summation, concatenation, or simple 1×1 convolution, it may increase the attention on less important objects, leading to information redundancy and affecting classification accuracy. To address this issue, we devise a KSMSA module, which employs varying large convolutional kernels and utilizes dynamic receptive fields to perceive information across different scales. Specifically, the dynamic receptive fields are realized through a spatial selective mechanism that efficiently weighs the features processed by a sequence of large convolutional kernels and subsequently integrates them spatially. The weights assigned to these kernels are dynamically determined based on the input, enabling the model to adaptively utilize different large kernels and adjust the receptive field for each spatial target as required. This approach effectively captures the subtle contextual nuances that distinguish various object categories, thereby enhancing the efficiency of feature fusion and bolstering the robustness of feature representation.

As shown in Fig. 4, the KSMSA module is composed of a large kernel selection and convolutional feature fusion and transformation layer. The large kernel selection module is embedded within two 1×1 convolutional layers that are used to enhance the network's representation ability. Its main steps are separation, fusion, and selection. For the jth stege, the large kernel selection module takes k feature maps $F_i \in \mathbb{R}^{C^j \times H^j \times W^j}$ ($i = 1, 2, \ldots, k$ and $j = 1, 2, 3, 4$) from MBM as input and apply convolution calculations with kernel sizes of 5 and 7, resulting in two features $\hat{U}_i$ and $\tilde{U}_i$ that have the same shape as F_i. Where C^j, H^j, and W^j represent the number of channels, spatial height, and width of the feature map at the current stage, respectively. This process can be formalized as

$$\hat{U}_i = \text{Conv}_{5\times5}(F_i) \tag{1}$$

$$\tilde{U}_i = \text{Conv}_{7\times7}(F_i) \tag{2}$$

where $\text{Conv}_{5\times5}(\cdot)$ and $\text{Conv}_{7\times7}(\cdot)$ are implemented using efficiently grouped convolutions, allowing for channel mixing within each spatial feature vector. The purpose of aggregation is to adaptively adjust the receptive field size based on content, with the fundamental idea being to use gates to control the information flow between the two convolutional kernel branches. To this end, the gating mechanism needs to integrate this information by concatenating these features, as follows:

$$U_i = \text{Concat}(\hat{U}_i, \tilde{U}_i) \tag{3}$$

where $\text{Concat}(\cdot, \cdot)$ represent concatenating operation.

To efficiently capture spatial relationships, parallel procedures incorporating global average pooling and max pooling are implemented, with their outputs subsequently concatenated to generate comprehensive spatial-wise statistical information. This methodology can be mathematically expressed as

$$S_i^{\text{avg}} = \text{AvgPool}(U_i) \tag{4}$$

$$S_i^{\text{max}} = \text{MaxPool}(U_i) \tag{5}$$

$$S_i = \text{Concat}(S_i^{\text{avg}}, S_i^{\text{max}}) \tag{6}$$

where $\text{AvgPool}(\cdot)$ and $\text{MaxPool}(\cdot)$ represent global average pooling and max pooling, respectively. Subsequently, the pooled features are converted into two spatial attention maps using convolutions, enabling information interaction across different spatial regions. A sigmoid activation function is employed to obtain selection masks for each decomposed large kernel's individual spatial components, guiding precise and adaptive selection. This process can be formalized as

$$z_i = \sigma(\text{Conv}(S_i)) \in \mathbb{R}^{2\times H^j \times W^j} \tag{7}$$

where $\sigma(\cdot)$ represents the sigmoid activation function. The features $\hat{U}_i$ and $\tilde{U}_i$ are reweighted using weight matrix z_i^1 and z_i^2 ($z_i = \text{Concat}(z_i^1, z_i^2)$), and summed to obtain the output

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feature V_i of the current scale. Additionally, to prevent gradient vanishing and gradient explosion, a residual connection is added. This process can be formalized as

$$V_i = \left(z_i^1 \cdot \hat{U}_i + z_i^2 \cdot \tilde{U}_i\right) + F_i. \tag{8}$$

Finally, the features from each branch that have undergone the large kernel selection operation are concatenated together, and then passed through a 1×1 convolution for channel transformation, completing the multibranch feature fusion to input into the MSEM of the next stage.

C. Learnable Multilevel Aggregation

CNNs excel at capturing shape features in their shallow layers and semantic features in deeper layers. However, traditional classification networks often solely rely on the latter layer for classification, overlooking the crucial role of shallow, low-level features. Notably, in the specific domain of RS scene classification, where texture prominence is diminished, shape information becomes paramount for accurate classification. While Transformers exhibit some capability in capturing shape variations, the pyramid-shaped structure of our network, with downsampling at each stage, inevitably leads to a weakening of the shape information. Furthermore, as the depth of the network increases, the loss of shape information becomes more pronounced. Consequently, we introduce the LMLA module, which aggregates shallow information features and strategically fuses them with original deep features to mitigate this issue.

Distinct from traditional pyramid feature aggregation relying on weighted averaging or max pooling for fusion, the LMLA incorporates a learnable Params ϵ in the fusion process, rendering the data-driven approach more tailored for RS scene classification. Specifically, as shown in the yellow module in the upper half of Fig. 3, the structure of LMLA involves downsampling the features from the three shallow stages to obtain feature maps of the same size. These features are then concatenated together, and through the SKConv [56] calculation, the fused feature $F^s \in \mathbb{R}^{C^4 \times H^4 \times W^4}$ of the utilized shallow information are derived. This process can be formalized as

$$F^s = \mathrm{Conv}_s\left(\mathrm{Concat}\left(\mathrm{DS}_1(F^1), \mathrm{DS}_2(F^2), \mathrm{DS}_3(F^3)\right)\right) \tag{9}$$

where Conv_s represents the SKConv calculation, $\mathrm{DS}_j(\dot{})$ represents the downsampling, and F^j represents the jth stage feature map. Then, the fused shallow feature F^s is integrated into the multiscale global feature F^4 extracted from the fourth stage through learnable Params ϵ. The specific process is shown as follows:

$$F = F^4 + \epsilon \cdot F^s. \tag{10}$$

The main consideration for incorporating shallow information is to supplement the information that may be missing from the deep semantic information. The setting of the ϵ should not be too large, as it may disrupt the deep semantic information. Therefore, the initial value should be set to a relatively small number. Finally, a cross-entropy loss function is used to optimize the network.

IV. Experiments

A. Dataset Introduction

Three public RS scene classification datasets are used to confirm the effectiveness of our proposed MBFNet, which are the event recognition in aerial (ERA) videos dataset [72], aerial image dataset (AID) [5], and University of California (UC) Merced (UCM) Land Use dataset [73]

1) Aerial Image Dataset: The AID dataset, released by Wuhan University, encompasses 30 scene categories and features 10 000 RS images, each with a size of 600 × 600 pixels. The number of images within each category ranges from 220 to 420, while the spatial resolution varies from approximately 0.5 to 8 m per pixel.

2) University of California (UC) Merced Land Use Dataset: The UCM dataset, published by the University of California Merced, comprises 2100 RS scene images categorized into 21 distinct scene categories. Each of these categories encompasses precisely 100 scene images, all of which have a standard size of 256 × 256 pixels. Moreover, each pixel in this dataset boasts a spatial resolution of 0.3 m in the red-green-blue (RGB) color space. This dataset is the most widely used in the RS classification community.

3) Event Recognition in Aerial: The ERA dataset was published by the German Aerospace Center. It contains 2864 RS scene images divided 25 into categories. The train and test datasets comprise 1473 and 1391 images with a size of 640 × 640, respectively. It covers 24 common event scenes, encompassing six major categories of disasters, traffic accidents, production activities, safety, sports, and social activities, as well as a non-event scene. This dataset is designed to have significant intraclass variability and interclass similarity, and includes event images under various conditions and scales.

B. Dataset Settings and Evaluation Metrics

For a fair comparison, consistent with previous works [11], [18], we conduct the experiments five times, randomly selecting training/testing samples to ensure reliable experimental outcomes. Subsequently, we report the average results and standard deviations. The training-testing ratios for the AID and UCM datasets are set to 20%–80% and 50%–50%, and 50%–50% and 80%–20%, respectively. To evaluate the classification performance, we select two widely used metrics: overall accuracy (OA) and confusion matrix (CM). As defined in 11, OA is calculated as the number of correctly classified images divided by the total number of testing images, providing an indication of the overall performance of a classification model

$$\mathrm{OA} = \frac{1}{N}\sum_{i=1}^{r} n_i \tag{11}$$

where N represents the total number of test samples, r represents the number of categories, and n_i represents the number of correctly predicted samples for the current category. The CM serves as a tool to dissect the intricacies of classification errors and the extent of confusion among various scene classes. Within the CM, each row represents the predicted class, while each column signifies the actual class.

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TABLE I
DETAILS TYPES OF DIFFERENT METHODS

Methods	Type	Venue	Year
VGG-16 [57]	GDF	ICLR	2015
ResNet-50 [19]	GDF	CVPR	2016
GoogLeNet [58]	GDF	TGRS	2017
CaffeNet [5]	GDF	TGRS	2017
MSCP [59]	MFA	TGRS	2018
MobileNet [60]	GDF	CVPR	2018
CapsNet [33]	MFA	RS	2019
SCCov [61]	MFA	TNNLS	2019
GBNet [7]	MFA	TGRS	2020
GPAS [62]	GDF	TGRS	2020
MG-CAP [63]	GDF	TIP	2020
MIDC-Net_CS [64]	GDF	TIP	2020
MS2AP [65]	GDF	Neurocomputing	2021
T2T-ViT [42]	TB	ICCV	2021
MLFF [66]	MFA	GRSL	2021
ACNet [38]	GDF	JSTARS	2021
MSA [14]	GDF	JSTARS	2021
RANet [67]	MFA	JSTARS	2021
EFPN [31]	MFA	TGRS	2021
RTRMM [32]	MFA	TGRS	2021
DFAGCN [30]	GCN	TNNLS	2021
ViT-B [39]	TB	ICLR	2021
EMTCAL [18]	MFA	TGRS	2022
PVT-V2 [68]	TB	CVM	2022
MPViT [15]	MFA	CVPR	2022
CGINet [69]	MFA	TGRS	2024
CGFNet [70]	MFA	TGRS	2024
GMFANet [71]	MFA	TGRS	2024
MBFNet			

TABLE II
PARAMS, FLOPS, OA RESULTS, AND STANDARD DEVIATIONS (%) OF DIFFERENT METHODS ON THE AID DATASET. THE BOLD INDICATES THE STATE-OF-THE-ART AND THE UNDERLINE INDICATES THE SECOND BEST

Methods	Parameters	FLOPs	Training Ratio	
			20%	50%
VGG-16 [57]	134.38M	15.47G	86.59±0.29	89.64±0.36
ResNet-50 [19]	26.10M	4.14G	92.39±0.23	94.69±0.31
GoogLeNet [58]	6.04M	1.58G	83.44±0.40	86.39±0.55
CaffeNet [5]	-	-	86.86±1.12	89.53±0.88
MSCP [59]	138.36	13.42G	92.21±0.17	95.56±0.18
MobileNet [60]	2.28M	0.41G	88.53±0.14	90.91±0.20
CapsNet [33]	-	-	91.63±0.19	94.74±0.17
SCCov [61]	-	-	93.12±0.25	96.10±0.16
GBNet [7]	-	-	92.20±0.23	95.48±0.12
GPAS [62]	-	-	-	95.28±0.14
MG-CAP [63]	-	-	93.34±0.18	96.12±0.12
MIDC-Net [64]	26.12M	2.89G	88.51±0.41	92.95±0.17
MS2AP [65]	27.49M	4.08G	92.19±0.31	94.82±0.23
T2T-ViT [42]	22.00M	6.10G	94.39±0.22	96.29±0.24
MLFF [66]	-	-	92.73±0.12	95.06±0.33
ACNet [38]	-	-	93.33±0.29	95.38±0.29
MSA [14]	-	-	93.53±0.21	96.01±0.43
RANet [67]	26.30M	6.96G	92.71±0.14	95.31±0.37
EFPN [31]	-	-	94.02±0.21	94.50±0.30
RTRMM [32]	0.82M	0.38	-	94.35±0.28
DFAGCN [30]	-	-	-	94.88±0.22
ViT-B [39]	85.63M	32.20G	93.74±0.27	95.84±0.29
EMTCAL [18]	27.30M	4.23G	94.69±0.14	96.41±0.23
PVT-V2 [68]	13.1M	2.10	93.52±0.35	96.27±0.14
CGINet [69]	26.10M	4.14G	95.35±0.14	97.02±0.23
CGFNet [70]	23.57M	4.13G	95.20±0.14	96.82±0.18
GMFANet [71]	23.11M	3.78G	95.74±1.35	97.06±0.74
MPViT [15]	20.23M	4.18G	94.61±0.09	96.58±0.13
MBFNet	22.09M	4.25G	**95.81**±0.13	**97.08**±0.17

C. Experimental Settings

We implement the experiment using PyTorch and MindSpore framework on CPU: E5-2609V4*2 @1.70 GHz; GPU: Nvidia GeForce RTX 2080Ti GPU*1; RAM: 128 GB; Ubuntu 20.04. We adopt the widely used MPViT [15] as the backbone network of MBFNet. Random horizontal flips and random rotation are used to augment the data during training. All input images are resized into 224 × 224 pixels. The number of iterations during training is 50 and the batch size is set to 64. The AdamW algorithm with weight decay set to 5e-2, learning rate set to 1e-4, and is selected as the optimizer. In addition, the learnable hyperparameter ϵ in 10 initialized to 0.1. The backbone network is initialized with the weight Params trained in ImageNet-1k [74].

D. Comparisons of State-of-the-Art Models

To thoroughly evaluate the classification performance of the proposed MBFNet, we conduct a comparative analysis of our method with several state-of-the-art models, including those based on global deep features (GDF), multilevel feature aggregation (MFA), Transformers (TB), and graph convolutional networks (GCNs). The comprehensive details of these comparative models are outlined in Table I, where all implementations are sourced from their respective official versions and the batch sizes are consistent with those used in this article.

1) Comparison on AID: The OA values of different methods on the AID Dataset are summarized in Table II. Among the various methods compared, the proposed MBFNet exhibits superior performance, attaining remarkably outstanding results. Specifically, when 20% of the scenes are utilized for training, our method achieves an OA of 95.81%, surpassing the second-best method (GMFANet) by 0.07%. Furthermore, with 50% of the scenes used for training, our MBFNet attains an OA of 97.08%, exceeding the second-best method, GMFANet, by 0.02%. Compared to the baseline, our method has improved by 1.20% and 0.50% on different training ratios, respectively.

When compared to GDF methods, such as VGGNet-16, MFA approaches (e.g., MSCP and CapsNet) exhibit better classification performance. This is attributed to the fact that the features generated through the aggregation strategy are more discriminative. Additionally, the performance of TB models surpasses that of traditional CNN models, further validating the Transformer model's prowess in mining global contextual information. While the aforementioned methods demonstrate commendable performance, our MBFNet achieves superior performance. The reasons behind this can be encapsulated as follows: First, our MBFNet leverages LMLA to integrate local information and its contextual relationships within RS scenes, resulting in a discriminative feature representation. Second, MBFNet employs KSMSA to effectively extract multiscale contextual information and long-range dependencies across different feature levels, thus enabling a comprehensive interpretation of RS scene contents.

In addition to the OA, we present the class-wise accuracy and CMs for various training ratios in Figs. 5 and 6. At a training ratio of 20%, as depicted the solid line in Fig. 5, our method exhibits significant enhancements in the "Church,"

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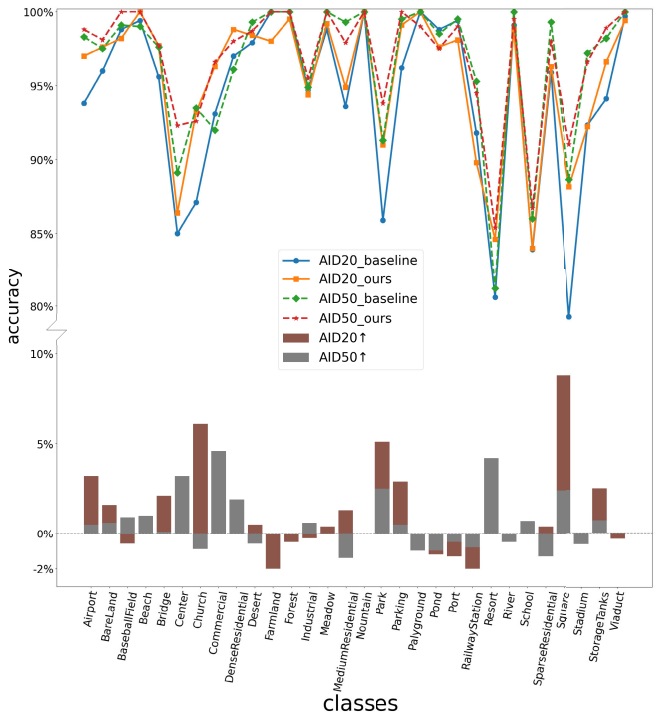


Fig. 5. Accuracy curves and the magnitude of performance improvement for each class on the AID dataset (AID20: AID dataset with 20% trainable data, AID50: AID dataset with 50% trainable data, ↑: performance improvement).

"Park," and "Square" categories, highlighting the effectiveness of the multiscale feature fusion approach. Analyzing the CMs of the baseline and our proposed method, as shown in Fig. 6(a) and (b), reveals a reduction in the misclassification of "Bare Land" as "Airport," enhancing the recognition accuracy for this class. This underscores the importance of supplementing shallow shape features and utilizing detailed texture information to distinguish effectively between these two similar categories. Furthermore, the mutual misclassifications between "Dense Residential" and "Resort" have significantly decreased in both directions, validating the effectiveness of the dynamic receptive field aggregation method employed in this article. For dense residential images, a smaller receptive field is crucial, while for relatively sparse images, a larger receptive field is paramount. The precise capture of image features has contributed to an improvement in the accuracy of the respective categories.

At a 50% training ratio, an analysis of the accuracy curves for each category (the intermittent line in Fig. 5) reveals a substantial increase in classification accuracy for the "Commercial" and "Resort" categories, along with notable improvements across most other categories. Comparing the CMs of the baseline and our proposed method [Fig. 6(b) and (c)], it becomes evident that there is a reduction in the number of "Square" instances incorrectly labeled as "Square," highlighting the importance of shape information. This distinction is effectively achieved through the utilization of shallow architectural shape information. Additionally, the "Commercial" category is no longer misclassified as "Center." This underscores the effectiveness of the KSMSA module, which employs varied receptive fields for different images and process features using appropriate receptive fields. This method ensures that crucial information is captured while preventing feature redundancy during multiscale fusion, resulting in an accurate and appropriate representation of the discriminative features of the images.

2) Comparison on UCM: Table III details the classification outcomes of various methods on the UCM dataset. Akin to the results on the AID dataset, it is evident that our MBFNet excels in all scenarios. Specifically, with 50% of the images used for training, MBFNet achieves an OA of 98.86%, surpassing the second-best method, GMFANet, by 0.06%. Similarly, when 80% of the images are utilized for training, MBFNet attains an OA of 99.76%, outperforming GMFANet by 0.10%. Compared to the baseline, our approach demonstrates improvements of 0.48% under varying training ratios. An intriguing finding is that MFA-based methods, notably EMTCAL and GMFANet, demonstrate significant superiority over GDF-based models, especially when benchmarked against traditional CNN models such as CaffeNet and VGG-16. This underscores the importance of considering the unique characteristics of RS scenes. In contrast to conventional CNN models, EMTCAL and GMFANet integrate attention mechanisms and transform-driven global features into their learning process, enabling the exploration of intricate contents and multiscale features in RS scenes. Additionally, our MBFNet's remarkable results reinforce this observation. Thanks to its multiscale feature extraction and multilevel aggregation mechanism, MBFNet comprehensively captures the rich and diverse land covers in RS scenes. These positive outcomes validate the effectiveness of our MBFNet.

Besides, the accuracy curves of each class and CMs under different training ratios are shown in Figs. 7 and 8. At a training ratio of 20%, as highlighted by the solid line in Fig. 7, the recognition accuracy of categories like "Forest," "Overpass," "River," and "Storagetanks" has reached 100%, while notable improvements have also been observed in "GolfCourse" and "MediumResidential." Detailed analysis of the CMs in Fig. 8(a) and (b) reveals that "River" is no longer misidentified as "Forest" or "Golf course," and "Forest" is correctly distinguished from "Agriculture." This underscores the efficacy of leveraging shallow shape information in this chapter, enabling precise differentiation between trees, grass, and rivers. However, the recognition accuracy of the "DenseResidential" category has declined, likely attributed to the uniform image scale and inconspicuous multiscale information present in all images of the "DenseResidential" test set, as depicted in Fig. 9. This has led to information redundancy within the proposed aggregation module, ultimately diminishing the classification effectiveness.

At a training ratio of 80%, as depicted by the intermittent line in Fig. 7, the MBFNet method demonstrates a notable enhancement, primarily manifested in the substantial boost to the accuracy of the "TennisCourt" category. By conducting a side-by-side comparison of the CMs [Fig. 8(c) and (d)], it becomes apparent that the model has successfully rectified the misclassifications of images that were previously erroneously labeled as "MediumResidential." This underscores the model's proficiency in effectively capturing intricate,

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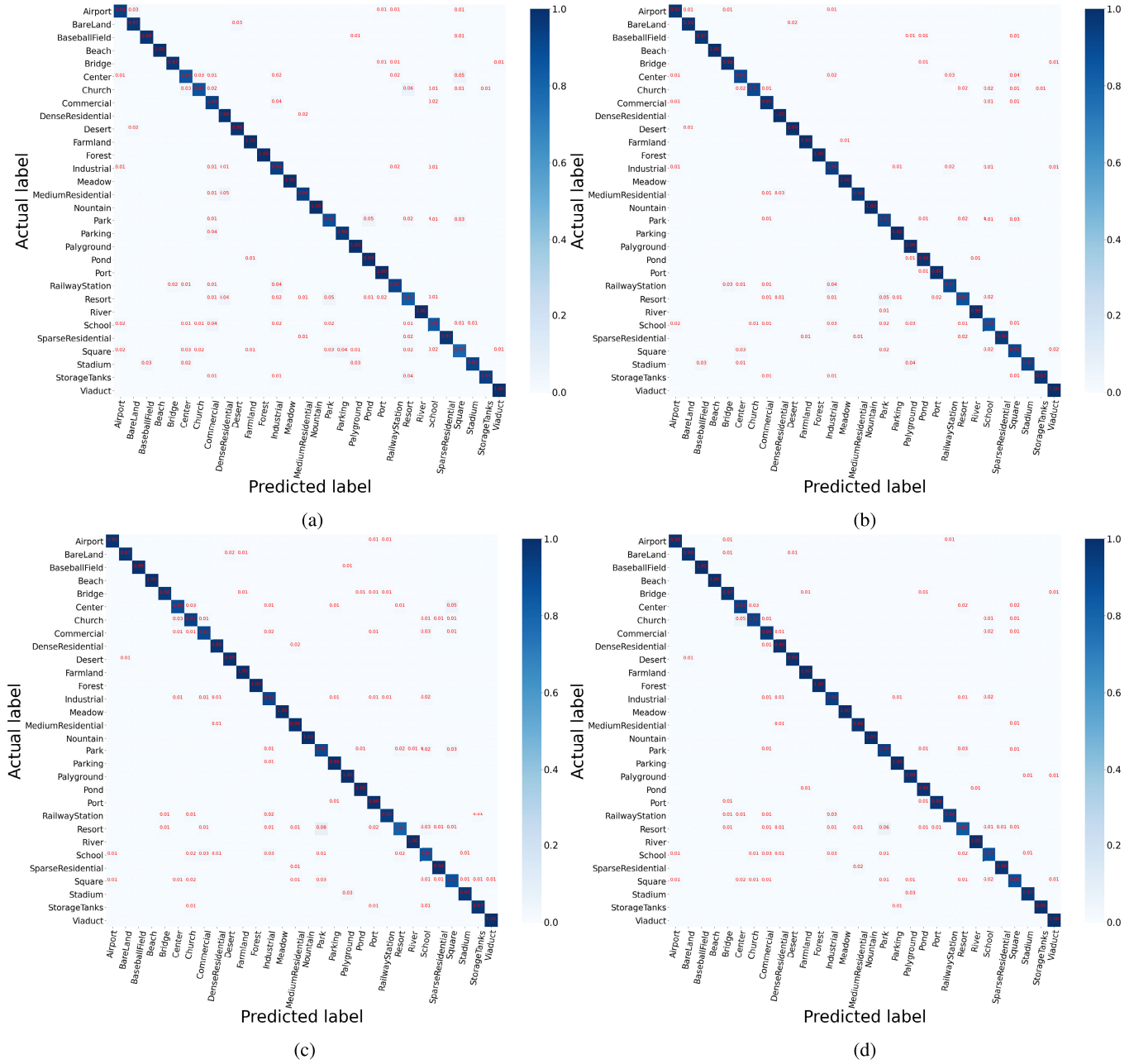


Fig. 6. CMs results on the AID dataset (AID20: AID dataset with 20% trainable data, AID50: AID dataset with 50% trainable data). (a) CM of baseline on AID20. (b) CM of MBFNet on AID20. (c) CM of baseline on AID50. (d) CM of MBFNet on AID50.

small-scale features within the images, contributing to its improved performance.

3) Comparison on ERA: The outcomes of different methods are concisely summarized in Table IV, revealing that our MBFNet method outperforms all others. For the ERA dataset, we adhered to the default partitioning of training and testing sets to compare traditional CNN models, encompassing ResNet [19], VGG [57], DenseNet [75], Inception [76], and MobileNet [60], alongside more contemporary other models such as ViT [39], Swin-ViT [40], MPViT [15], CGINet [69], CGFNet [70], and GMFANet [71]. As clearly depicted in Table IV, the method introduced in this article demonstrates remarkable performance on the ERA dataset, achieving a notable OA enhancement of 3.17% from 63.98% to 67.15% when compared to the baseline. Furthermore, its superior performance surpasses that of all other models, strongly underscoring the effectiveness of the proposed method in significantly boosting the accuracy of RS scene classification. Specifically, it surpasses the second-best method, GMFANet [71], by a substantial margin of 0.62%.

From the classification accuracy curves and the substantial performance enhancements per class depicted in Fig. 10, it is evident that our MBFNet method has significantly boosted the accuracy of numerous classes, with particularly remarkable improvements observed in the "Boating," "Landslide," and "Parade Protest" categories. This underscores the efficacy of the proposed aggregation module in effectively integrating multiscale features, mitigating semantic discrepancies across

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TABLE III
OA RESULTS AND STANDARD DEVIATIONS (%) OF DIFFERENT METHODS ON THE UCM DATASET

Methods	Training Ratio	
	50%	80%
VGG-16 [57]	94.14±0.69	95.21±1.20
ResNet-50 [19]	-	98.46±0.17
GoogLeNet [58]	92.70±0.60	94.31±0.89
CaffeNet [5]	93.98±0.57	95.02±0.72
MSCP [59]	-	98.40±0.34
MobileNet [60]	92.17±0.19	96.33±0.29
CapsNet [33]	95.33±0.18	98.81±0.22
SCCov [61]	-	99.05±0.25
GBNet [7]	97.05±0.19	98.57±0.48
GPAS [62]	-	98.20±0.32
MG-CAP [63]	-	99.00±0.10
MIDC-Net_CS [64]	95.41±0.40	97.40±0.48
MS2AP [65]	98.38±0.28	99.01±0.11
T2T-ViT [42]	97.79±0.20	99.10±0.32
MLFF [66]	97.99±0.26	99.37±0.15
MSA [14]	97.80±0.33	98.96±0.21
RANet [67]	97.80±0.19	99.27±0.24
EFPN [31]	96.19±0.13	99.14±0.22
RTRMM [32]	-	96.76±0.31
DFAGCN [30]	-	98.48±0.42
ViT-B [39]	97.83±0.16	98.95±0.24
EMTCAL [18]	98.67±0.16	99.57±0.28
PVT-V2 [68]	97.94±0.44	98.86±0.38
CGINet [69]	98.16±0.24	99.48±0.13
CGFNet [70]	98.48±0.42	99.57±0.23
GMFANet [71]	98.80±0.82	99.66±0.12
MPViT [15](baseline)	98.38±0.10	99.28±0.13
MBFNet(ours)	**98.86**±0.11	**99.76**±0.08

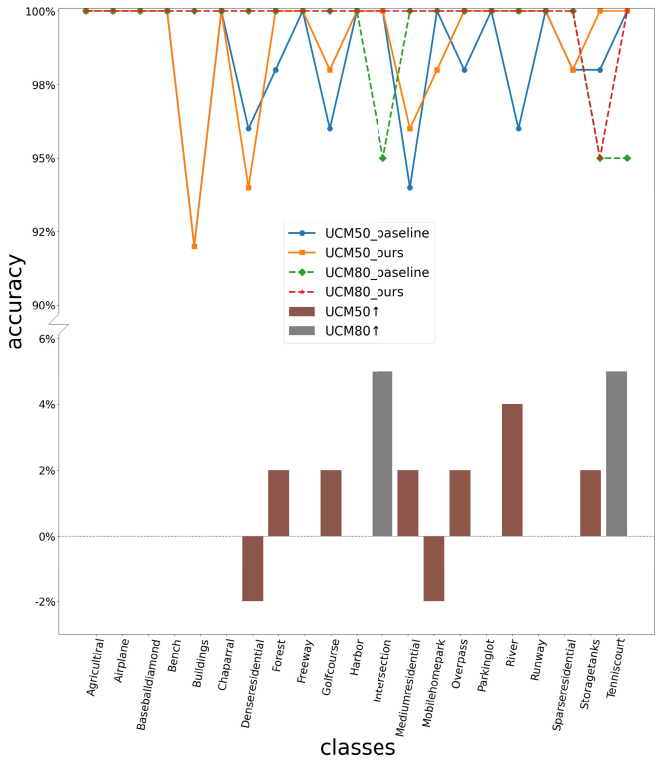


Fig. 7. Accuracy curves and the magnitude of performance improvement for each class on the UCM dataset (UCM50: UCM dataset with 50% trainable data, UCM80: UCM dataset with 80%trainable data).

varying scales, and consequently enhancing the accuracy of image scenes characterized by substantial scale variations.

Furthermore, examining the CMs presented in Fig. 11, which compare the baseline method with the one proposed

TABLE IV
OA RESULTS AND STANDARD DEVIATIONS (%) OF DIFFERENT METHODS ON THE ERA DATASET

Methods	OA
Swin-T [40]	64.20±0.19
ViT-B [39]	63.58±0.41
NASNet-L [77]	60.24±0.55
DenseNet-121 [75]	61.71±0.33
DenseNet-169 [75]	60.56±0.28
DenseNet-201 [75]	62.33±0.52
ResNet-50 [19]	57.32±0.39
ResNet-101 [19]	55.30±0.25
ResNet-152 [19]	56.12±0.66
MobileNet [60]	61.29±0.21
Inception-v3 [76]	62.03±0.37
VGG-16 [57]	51.92±0.28
VGG-19 [57]	49.71±0.49
CGINet [69]	64.29±0.52
CGFNet [70]	65.10±0.83
GMFANet [71]	66.53±0.76
MPViT [15]	63.98±0.46
MBFNet(ours)	**67.15**±0.32

in this article, reveals a decrease in the misclassifications of "Boating" as "Swimming." This finding indicates that within image contexts featuring extensive water backgrounds, the method effectively discounts irrelevant background details within the enlarged receptive field. This emphasizes the importance of augmenting shallow information in our approach, which enables a more precise utilization of shapes and intricate features. Additionally, for the frequently confounded categories of "Parade Protest" and "Religious Activity," our method demonstrates superior discrimination capabilities, markedly reducing the instances of misclassifications between these two classes and substantially enhancing the accuracy of predictions.

E. Ablation Study

1) Computational Complexity and Number of Params: To better explain the computational complexity and the number of Params of our model, Table II presents the floating-point operations (FLOPs) and Params for our method, baseline (MPViT), and other comparison algorithms. Our MBFNet incorporates KSMSA and LMLA modules to capture diverse scale and level features. Therefore, compared to the baseline, our method incurs only a minor increase of approximately 1.86 M Params and up to 0.07 G FLOPs. Despite the slight increase in Params and computational complexity, MBFNet demonstrates significant performance enhancements, and this increase is not significant compared to other RS scene classification methods.

2) Evaluate the Statistical Significance of the Classifier: Statistical significance tests are employed to assess whether notable differences exist between the baseline MPViT and the proposed MBFNet. For this purpose, we selected the paired samples t-test to conduct experiments on the ERA and AID50 (AID with 50% training data) datasets. Initially, a null hypothesis was formulated: there is no statistically significant difference between MBFNet and MPViT. Subsequently, we initialized both models using identical random seeds, then altered the seeds and independently repeated the experiment eight times on both the ERA and AID50 datasets. The OA

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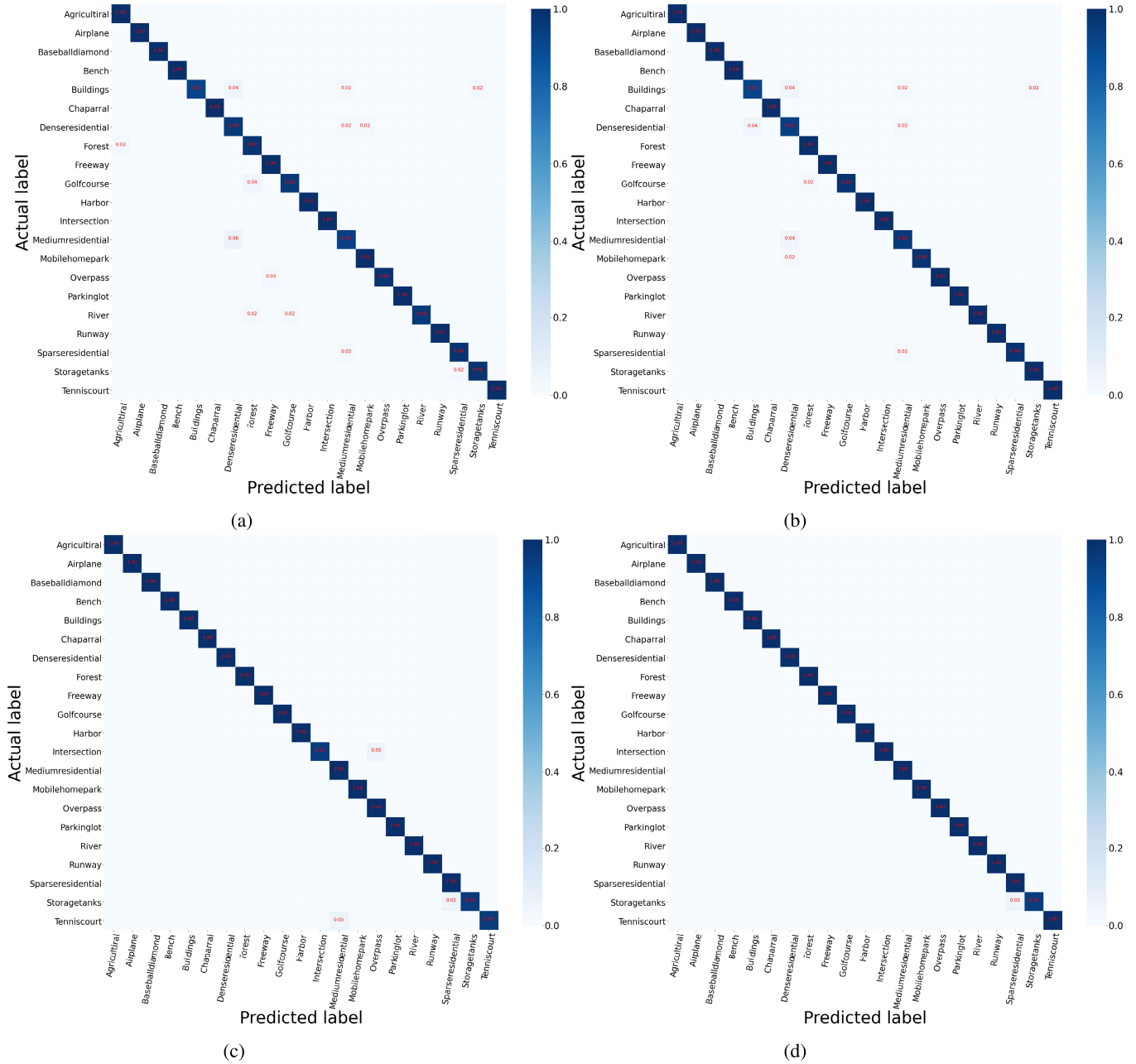


Fig. 8. CMs results on the UCM dataset (UCM50: UCM dataset with 50% trainable data, UCM80: UCM dataset with 80% trainable data). (a) CM of baseline on UCM50. (b) CM of MBFNet on UCM50. (c) CM of baseline on UCM80. (d) CM of MBFNet on UCM80.

values calculated by MBFNet and MPViT are presented in Table V. The results of the paired t-test yielded probabilities of $p = 3.80e^{-5}$ and $p = 0.0066$, both of which are significantly below the threshold of 0.05. Consequently, we reject the null hypothesis and conclude that there is a statistically significant difference between MBFNet and MPViT.

3) Effectiveness of Each Modules: To validate the efficacy of the modules introduced in MBFNet, ablation studies were conducted using the ERA dataset. Initially, we assessed the performance of the proposed LMLA module. As shown in Table VI, incorporating shallow-level information into deep-level features resulted in a modest yet noticeable improvement of 0.66% in classification accuracy (Baseline + LMLA). Subsequently, we evaluated the effectiveness of the KSMSA module. The results indicate that integrating the LMLA module led to a substantial increase in classification accuracy by 2.51% [Baseline + LMLA + KSMSA (ours)]. These findings

TABLE V

OA RESULTS OF THE BASELINE AND OUR MBFNET ON THE ERA AND AID WITH 50% TRAINING DATA DATASET.(1, 2, 3, AND 4 INDICATE THE RESPECTIVE EXPERIMENT NUMBERS CONDUCTED UNDER DIFFERENT RANDOM SEEDS, THE THIRD ROW: BASELINE RESULT, THE FOURTH ROW: MBFNET RESULT)

ERA				AID50			
1	**2**	**3**	**4**	**1**	**2**	**3**	**4**
63.28	64.10	64.24	63.36	96.60	96.41	96.55	96.76
67.43	66.68	67.35	66.84	97.19	97.27	96.93	96.88

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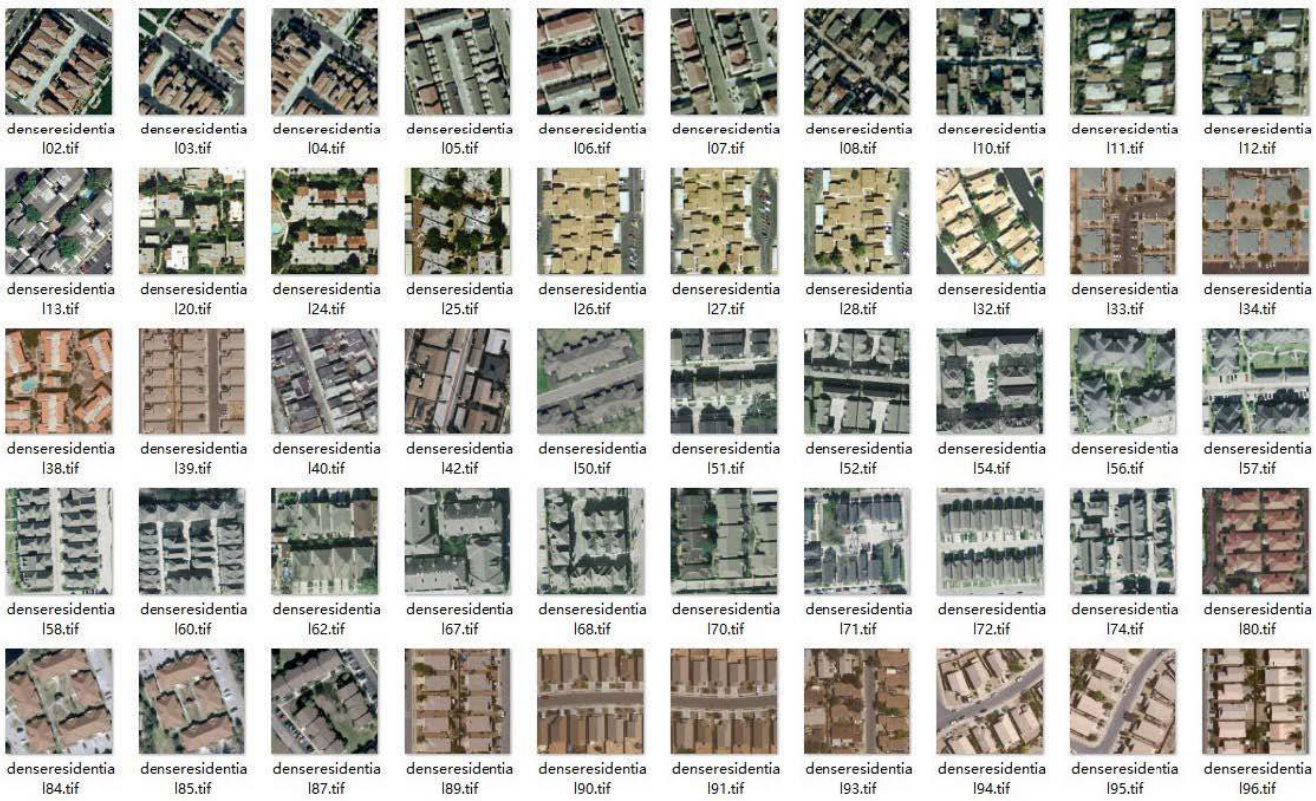


Fig. 9. Images sampled from the "dense residential" of test datasets.

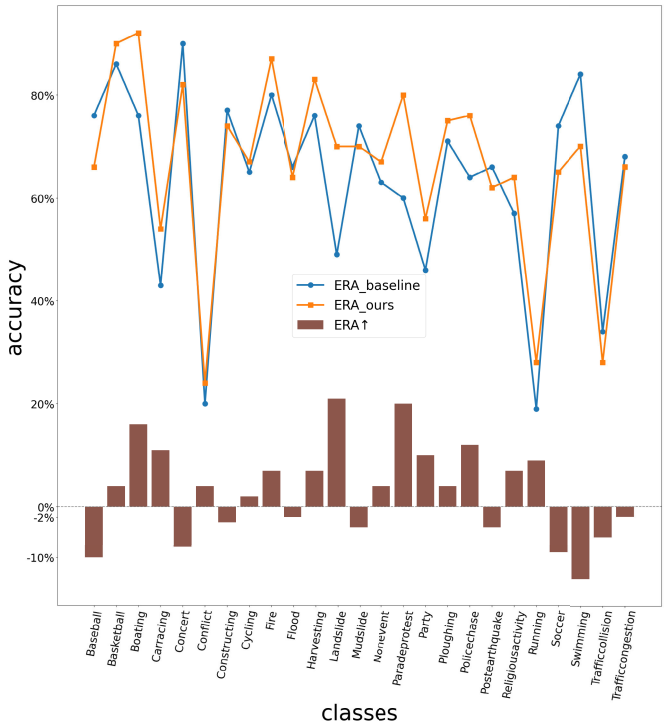


Fig. 10. Accuracy curves and the magnitude of performance improvement for each class on the ERA dataset.

confirm that both proposed modules significantly enhance the performance of RS scene classification.

To further validate the necessity of the KSMSA module, we conducted a comparative analysis between KSMSA and alternative methods for aggregating multiscale features. Our findings indicate that straightforward approaches, such as averaging (Aggregation with Avg) or concatenating (Aggregation with Concat) features across scales followed by a 1×1 convolution, resulted in suboptimal performance. These methods not only led to a decline in classification accuracy but also performed worse than the baseline. This highlights the limitations of conventional aggregation techniques in fusing features from CNN and Transformer multiscale branches. The root cause may lie in the fundamental differences between how these two network architectures process features, combined with significant semantic disparities across scales. When features are merged without proper distinction, it inadvertently reinforces feature representations across all regions, leading to considerable information redundancy. This redundancy diminishes the focus on salient features and severely compromises their discriminative power. Consequently, this reinforces the efficacy of the proposed KSMSA module, which utilizes dynamic receptive fields to process and aggregate features at different scales while aligning their semantic differences.

We also experimented with enhancing the kernel selection mechanism by integrating a smaller kernel branch specifically designed to handle finer-grained features (KSMSA + 3×3 conv). Surprisingly, although the inclusion of the small kernel branch resulted in a slight decline in classification accuracy, the overall performance still exceeded that of the baseline. This finding underscores the effectiveness of our kernel selection strategy in amalgamating multiscale features. However, the observed decrease in accuracy after incorporating smaller kernels might be attributed to the fact that,

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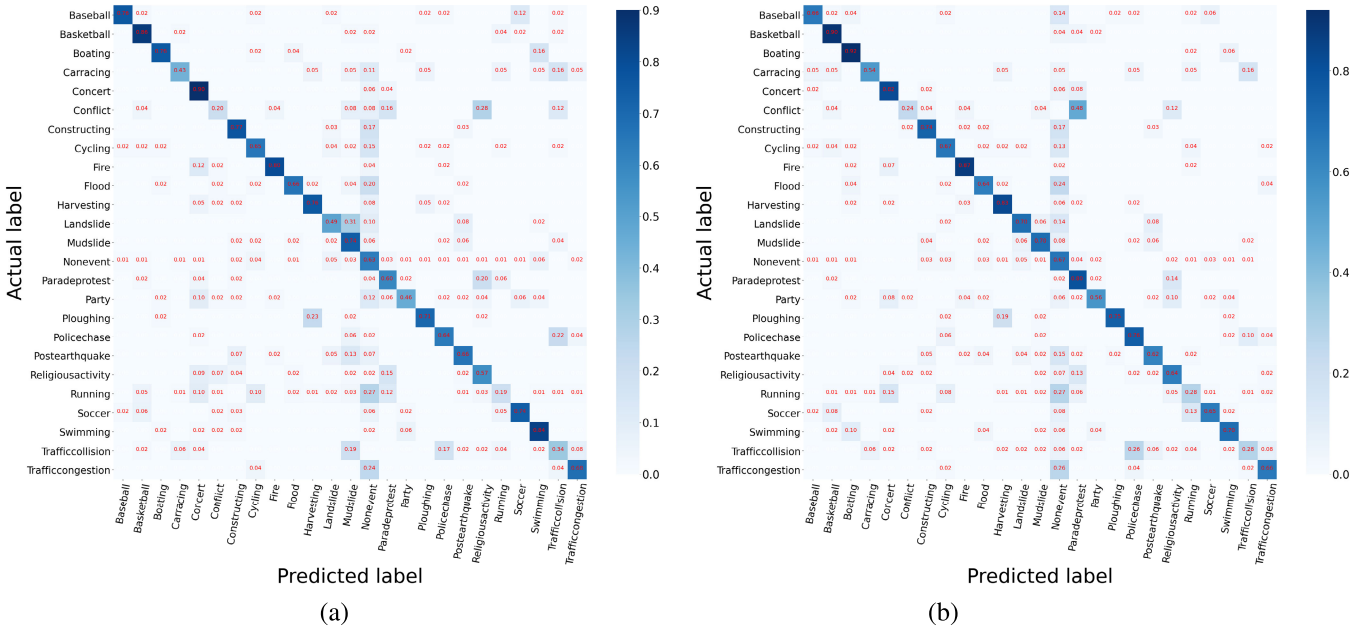


Fig. 11. CMs results on the ERA dataset. (a) CM of baseline. (b) CM of the proposed MBFNet.

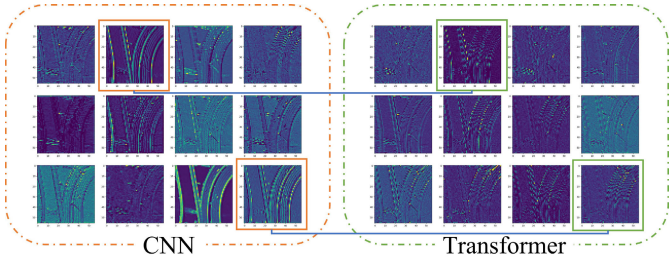


Fig. 12. Feature maps of shallow features extracted by CNN and Transformer, where the input image is the same as Fig. 2.

TABLE VI
OA RESULTS OF ABLATION STUDIES

Models	OA
Baseline	63.98±0.46
Baseline + LMLA	64.64±0.51
Aggregation with Avg	40.85±0.21
Aggregation with Concat	43.08±0.39
KSMSA + 3×3 conv	64.19±0.47
LMLA with Transformer	64.95±0.19
Baseline + LMLA + KSMSA (ours)	**67.15**±0.32

as previously demonstrated by RepLKNet [78], larger kernels are inherently more effective at processing even small-scale features with remarkable efficiency. Thus, the addition of smaller kernels could inadvertently introduce feature redundancy, which might detract from the model's discriminative power and ultimately compromise classification accuracy.

In the context of the LMLA module, there is a choice to be made between aggregating features from the CNN branch or the Transformer branch. To gain insight into this decision, we conducted a visual analysis of the feature maps generated by the shallow layers of both the CNN and the Transformer. As evident from Fig. 12, a closer look reveals that the CNN network's shallow layers capture more intricate shapes and sharper edge contours, which are valuable in enhancing the shape information present in deep features. To empirically support this observation, we performed ablation experiments (refer to the "LMLA with Transformer" row in Table VI). The experimental outcomes underscore that during multilevel fusion, aggregating features from the CNN branch yields a notable improvement in classification accuracy, outperforming the accuracy achieved by aggregating Transformer branch features by a margin of 2.20%.

V. CONCLUSION

In this article, we propose to combine multiscale and multilevel features within a multibranch fusion-based feature enhance framework called MBFNet for accurate RS scene classification. To this end, we develop a KSMSA to efficiently process the diverse scale features emanating from these parallel branches. By selecting different convolution kernels, a dynamic receptive field is established to adaptively process features of different scales, reducing semantic differences to achieve effective aggregation of multiscale features. Traditional classification networks often solely rely on the latter layers for classification, overlooking the crucial role of shallow, low-level features. To address this, we introduce an LMLA module that integrates multilevel information and dynamically incorporates it into deep features, enabling more comprehensive feature fusion. Benefiting from KSMSA and LMLA, the proposed MBFNet improves the discriminability of features, thereby enhancing classification performance. The compelling experimental results counted on three widely adopted datasets clearly demonstrate the efficacy of our MBFNet in facilitating RS scene classification tasks.

While the proposed MBFNet method demonstrates promising classification results, there remains potential for enhancement. First, we employ the MPViT [15] model pretrained on natural images to directly learn multiscale features from RS

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scenes. While this approach is feasible, the general-purpose framework may neglect the unique characteristics inherent to RS scenes. Consequently, one of our future directions involves developing an RS-oriented framework similar to RASNet [79] for resolution-agnostic RS scene classification to more effectively capture these multiscale features. Additionally, to prevent an increase in model complexity, the learnable Params utilized in LMLA is kept simple, which may constrain the model's performance. Therefore, another focal point for future work is the innovation of more effective methods for multilevel feature fusion.

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